Experiment No. 04 – Performance Evaluation Metrics for Data Models (Supervised)

AIM: Implement and explore performance evaluation metrics for Data Models (Supervised)

Theory

Classification Metrics

Metric	Formula	Evaluation Focus
Accuracy	ACC = (TP + TN) / (TP + TN + FP + FN)	Overall effectiveness of a classifier
Error Rate	ERR = (FP + FN) / (TP + TN + FP + FN)	Classification error
Precision	PRC = TP / (TP + FP)	Class agreement of data labels with positive labels given by the classifier
Sensitivity (Recall)	SNS = TP / (TP + FN)	Effectiveness of a classifier to identify positive labels
Specificity	SPC = TN / (TN + FP)	How effectively a classifier identifies negative labels
F_1 Score	$F_1 = 2 \cdot (PRC \cdot SNS) / (PRC + SNS)$	Combination of precision and sensitivity in a single metric
Geometric Mean	$GM = \sqrt{(SNS \cdot SPC)}$	Combination of sensitivity and specificity in a single metric

- **False Positive Rate (α)** = Type I error = 1 − Specificity = FP / (FP + TN)
- **False Negative Rate (β)** = Type II error = 1 − Sensitivity = FN / (TP + FN)
- **Power** = Sensitivity = 1 − β

Regression Metrics

Metric	Formula	Explanation
Pearson Correlation (r)	$r = \Sigma(x-\bar{x})(y-\bar{y}) / \sqrt{[\Sigma(x-\bar{x})^2 \cdot \Sigma(y-\bar{y})^2]}$	Strength of association between predicted and observed values
R^2 (Coefficient of Determination)	$R^2 = 1 - SS_res / SS_tot$	Fraction of variance explained by the model
Mean Squared Error (MSE)	$MSE = (1/n)\ \Sigma(y_i - \hat{y}_i)^2$	Most common loss function for regression
Root Mean Squared Error (RMSE)	$RMSE = \sqrt{MSE}$	Error rate of a regression model

Metric	Formula	Explanation
Mean Absolute Error (MAE)	MAE = (1/n) Σ	$y_i - \hat{y}_i$
MAPE	MAPE = (100%/n) Σ	$y_i - \hat{y}_i$

Dataset

Online Retail Dataset – Transactional data from a UK-based online retailer.

```
# Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression, LinearRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import (
    confusion_matrix, accuracy_score, precision_score, recall_score,
    f1_score, roc_curve, roc_auc_score, classification_report,
    mean_squared_error, mean_absolute_error, r2_score
)
from scipy.stats import pearsonr
import warnings
warnings.filterwarnings('ignore')

# Load Dataset
df = pd.read_csv('data_online_retail.csv', encoding='unicode_escape')
print(f"Shape: {df.shape}")
df.head()
```

```
Shape: (541909, 8)
```


```
{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"InvoiceNo","rawType":"object","type":"string"},
{"name":"StockCode","rawType":"object","type":"string"},
{"name":"Description","rawType":"object","type":"string"},
{"name":"Quantity","rawType":"int64","type":"integer"},
{"name":"InvoiceDate","rawType":"object","type":"string"},
{"name":"UnitPrice","rawType":"float64","type":"float"},
{"name":"CustomerID","rawType":"float64","type":"float"},
{"name":"Country","rawType":"object","type":"string"}],"ref":"41b170a6
-834c-4719-9ad9-8ff6026d837d","rows":[["0","536365","85123A","WHITE
HANGING HEART T-LIGHT HOLDER","6","12/1/2010
8:26","2.55","17850.0","United Kingdom"],["1","536365","71053","WHITE
METAL LANTERN","6","12/1/2010 8:26","3.39","17850.0","United
Kingdom"],["2","536365","84406B","CREAM CUPID HEARTS COAT
HANGER","8","12/1/2010 8:26","2.75","17850.0","United Kingdom"],
```

["3","536365","84029G","KNITTED UNION FLAG HOT WATER BOTTLE","6","12/1/2010 8:26","3.39","17850.0","United Kingdom"],
["4","536365","84029E","RED WOOLLY HOTTIE WHITE HEART.","6","12/1/2010 8:26","3.39","17850.0","United Kingdom"]],"shape":
{"columns":8,"rows":5}}

```
# Data Preprocessing
df = df.dropna(subset=['CustomerID'])
df = df[(df['Quantity'] > 0) & (df['UnitPrice'] > 0)]
df['TotalPrice'] = df['Quantity'] * df['UnitPrice']
print(f"Cleaned shape: {df.shape}")
df.info()
```

```
Cleaned shape: (397884, 9)
<class 'pandas.core.frame.DataFrame'>
Index: 397884 entries, 0 to 541908
Data columns (total 9 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   InvoiceNo    397884 non-null  object
 1   StockCode    397884 non-null  object
 2   Description  397884 non-null  object
 3   Quantity     397884 non-null  int64
 4   InvoiceDate  397884 non-null  object
 5   UnitPrice    397884 non-null  float64
 6   CustomerID   397884 non-null  float64
 7   Country      397884 non-null  object
 8   TotalPrice   397884 non-null  float64
dtypes: float64(3), int64(1), object(5)
memory usage: 30.4+ MB
```

Part 1: Classification Metrics

Task: Predict whether a transaction is from the **United Kingdom** or not (binary classification).

We will evaluate the classifier using: Accuracy, Error Rate, Precision, Sensitivity (Recall), Specificity, F_1 Score, Geometric Mean, and ROC-AUC.

```
# Prepare Classification Data
# Binary target: 1 = United Kingdom, 0 = Other
df['IsUK'] = (df['Country'] == 'United Kingdom').astype(int)

# Features
X_clf = df[['Quantity', 'UnitPrice', 'TotalPrice']]
y_clf = df['IsUK']

# Train-Test Split
X_train_c, X_test_c, y_train_c, y_test_c = train_test_split(
```

```
    X_clf, y_clf, test_size=0.3, random_state=42, stratify=y_clf
)

# Scale features
scaler = StandardScaler()
X_train_cs = scaler.fit_transform(X_train_c)
X_test_cs = scaler.transform(X_test_c)

print(f"Training set: {X_train_cs.shape[0]} samples")
print(f"Test set    : {X_test_cs.shape[0]} samples")
print(f"\nClass distribution:\n{y_clf.value_counts()}")
```

```
Training set: 278518 samples
Test set    : 119366 samples

Class distribution:
IsUK
1    354321
0     43563
Name: count, dtype: int64
```

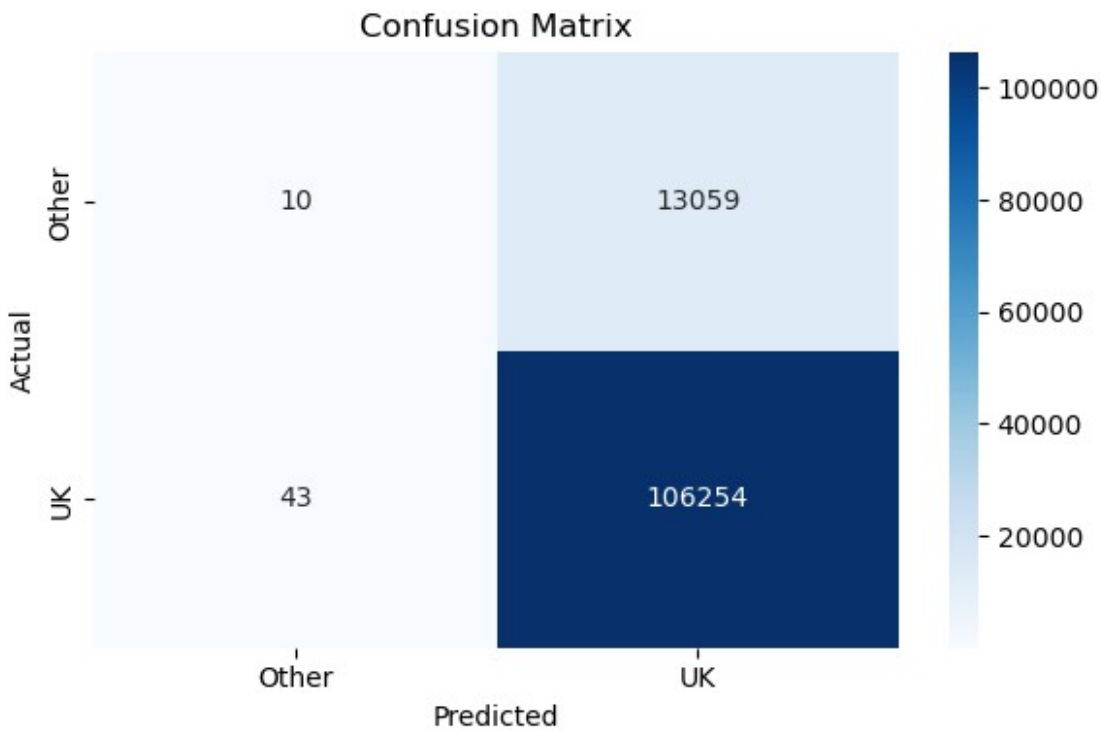
```
# Train Logistic Regression Classifier
clf = LogisticRegression(random_state=42, max_iter=1000)
clf.fit(X_train_cs, y_train_c)

# Predictions
y_pred_c = clf.predict(X_test_cs)
y_prob_c = clf.predict_proba(X_test_cs)[:, 1]

print("Classification Report:\n")
print(classification_report(y_test_c, y_pred_c, target_names=['Other',
'UK']))
```

```
Classification Report:

              precision    recall  f1-score   support

       Other       0.19      0.00      0.00     13069
          UK       0.89      1.00      0.94    106297

    accuracy                           0.89    119366
   macro avg       0.54      0.50      0.47    119366
weighted avg       0.81      0.89      0.84    119366
```

```
# Confusion Matrix
cm = confusion_matrix(y_test_c, y_pred_c)
TN, FP, FN, TP = cm.ravel()

print("Confusion Matrix:")
print(cm)
```

```python
print(f"\nTP = {TP}, TN = {TN}, FP = {FP}, FN = {FN}")

# Plot Confusion Matrix
plt.figure(figsize=(6, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Other', 'UK'], yticklabels=['Other', 'UK'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.tight_layout()
plt.show()
```

```
Confusion Matrix:
[[    10  13059]
 [    43 106254]]

TP = 106254, TN = 10, FP = 13059, FN = 43
```



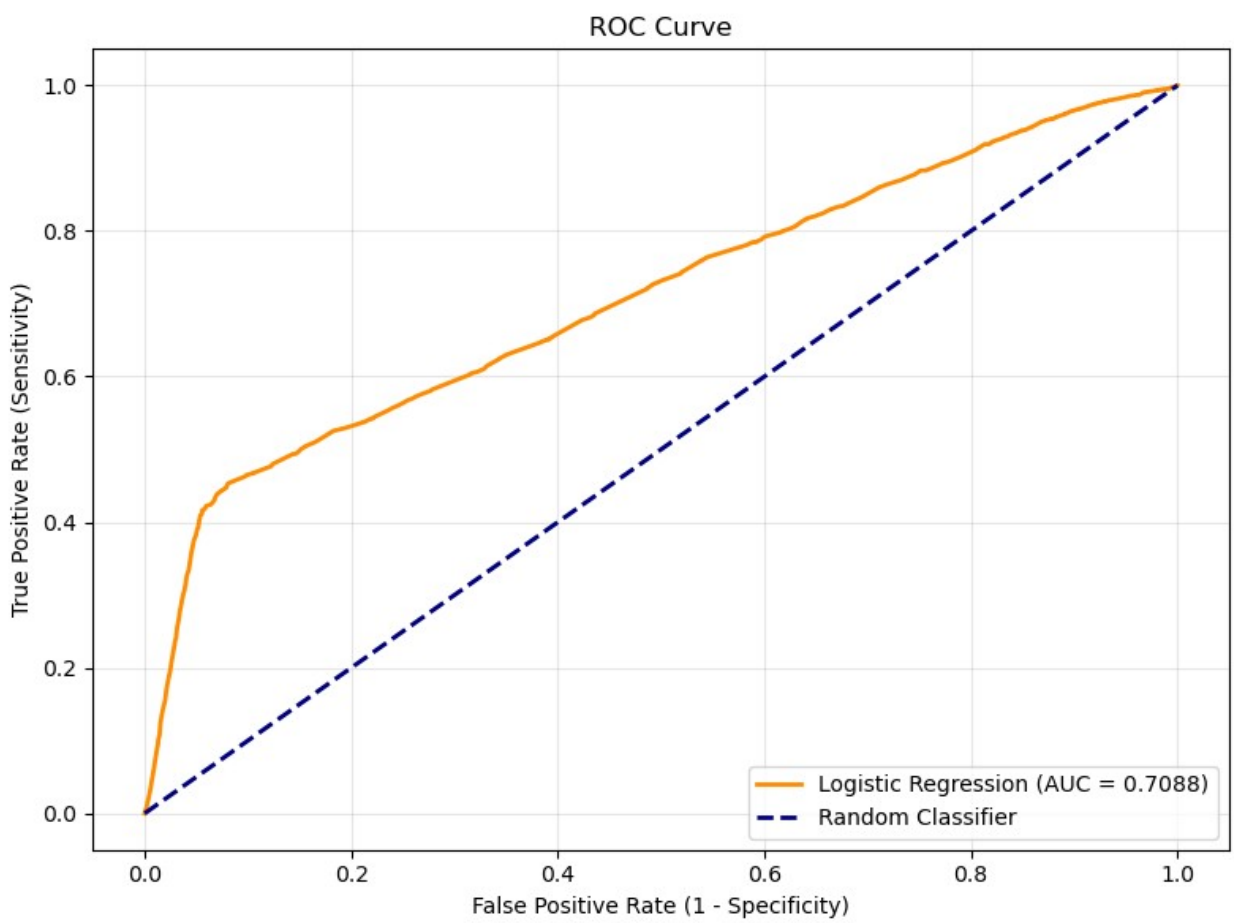

```python
# Compute All Classification Metrics
accuracy    = (TP + TN) / (TP + TN + FP + FN)
error_rate  = (FP + FN) / (TP + TN + FP + FN)
precision   = TP / (TP + FP)
sensitivity = TP / (TP + FN)        # Recall
specificity = TN / (TN + FP)
f1          = 2 * (precision * sensitivity) / (precision +
sensitivity)
```

```python
geometric_mean = np.sqrt(sensitivity * specificity)
fpr_val     = FP / (FP + TN)        # False Positive Rate (Type I
error)
fnr_val     = FN / (TP + FN)        # False Negative Rate (Type II
error)
power       = sensitivity            # Power = 1 - β

roc_auc     = roc_auc_score(y_test_c, y_prob_c)

print("=" * 50)
print("     CLASSIFICATION EVALUATION METRICS")
print("=" * 50)
print(f"  Accuracy              : {accuracy:.4f}")
print(f"  Error Rate            : {error_rate:.4f}")
print(f"  Precision             : {precision:.4f}")
print(f"  Sensitivity (Recall)  : {sensitivity:.4f}")
print(f"  Specificity           : {specificity:.4f}")
print(f"  F1 Score              : {f1:.4f}")
print(f"  Geometric Mean        : {geometric_mean:.4f}")
print(f"  False Positive Rate α : {fpr_val:.4f}")
print(f"  False Negative Rate β : {fnr_val:.4f}")
print(f"  Power (1 - β)         : {power:.4f}")
print(f"  ROC-AUC               : {roc_auc:.4f}")
print("=" * 50)

# Verify with sklearn
print("\n--- Verification with sklearn ---")
print(f"  Accuracy  (sklearn): {accuracy_score(y_test_c,
y_pred_c):.4f}")
print(f"  Precision (sklearn): {precision_score(y_test_c,
y_pred_c):.4f}")
print(f"  Recall    (sklearn): {recall_score(y_test_c,
y_pred_c):.4f}")
print(f"  F1 Score  (sklearn): {f1_score(y_test_c, y_pred_c):.4f}")
```

```
==================================================
     CLASSIFICATION EVALUATION METRICS
==================================================
  Accuracy              : 0.8902
  Error Rate            : 0.1098
  Precision             : 0.8905
  Sensitivity (Recall)  : 0.9996
  Specificity           : 0.0008
  F1 Score              : 0.9419
  Geometric Mean        : 0.0277
  False Positive Rate α : 0.9992
  False Negative Rate β : 0.0004
  Power (1 - β)         : 0.9996
  ROC-AUC               : 0.7088
==================================================
```

```
--- Verification with sklearn ---
  Accuracy  (sklearn): 0.8902
  Precision (sklearn): 0.8905
  Recall    (sklearn): 0.9996
  F1 Score  (sklearn): 0.9419
```

```
# Plot ROC Curve
fpr, tpr, thresholds = roc_curve(y_test_c, y_prob_c)

plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'Logistic
Regression (AUC = {roc_auc:.4f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--',
label='Random Classifier')
plt.xlabel('False Positive Rate (1 - Specificity)')
plt.ylabel('True Positive Rate (Sensitivity)')
plt.title('ROC Curve')
plt.legend(loc='lower right')
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



Part 2: Regression Metrics

Task: Predict **UnitPrice** from Quantity and TotalPrice using Linear Regression.

We will evaluate the model using: Pearson Correlation (r), R^2, MSE, RMSE, RMSRE, MAE, and MAPE.

```
# Prepare Regression Data
X_reg = df[['Quantity', 'TotalPrice']]
y_reg = df['UnitPrice']

# Train-Test Split
X_train_r, X_test_r, y_train_r, y_test_r = train_test_split(
    X_reg, y_reg, test_size=0.3, random_state=42
)

# Train Linear Regression Model
reg = LinearRegression()
reg.fit(X_train_r, y_train_r)

# Predictions
y_pred_r = reg.predict(X_test_r)

print(f"Training samples: {X_train_r.shape[0]}")
print(f"Test samples    : {X_test_r.shape[0]}")
```

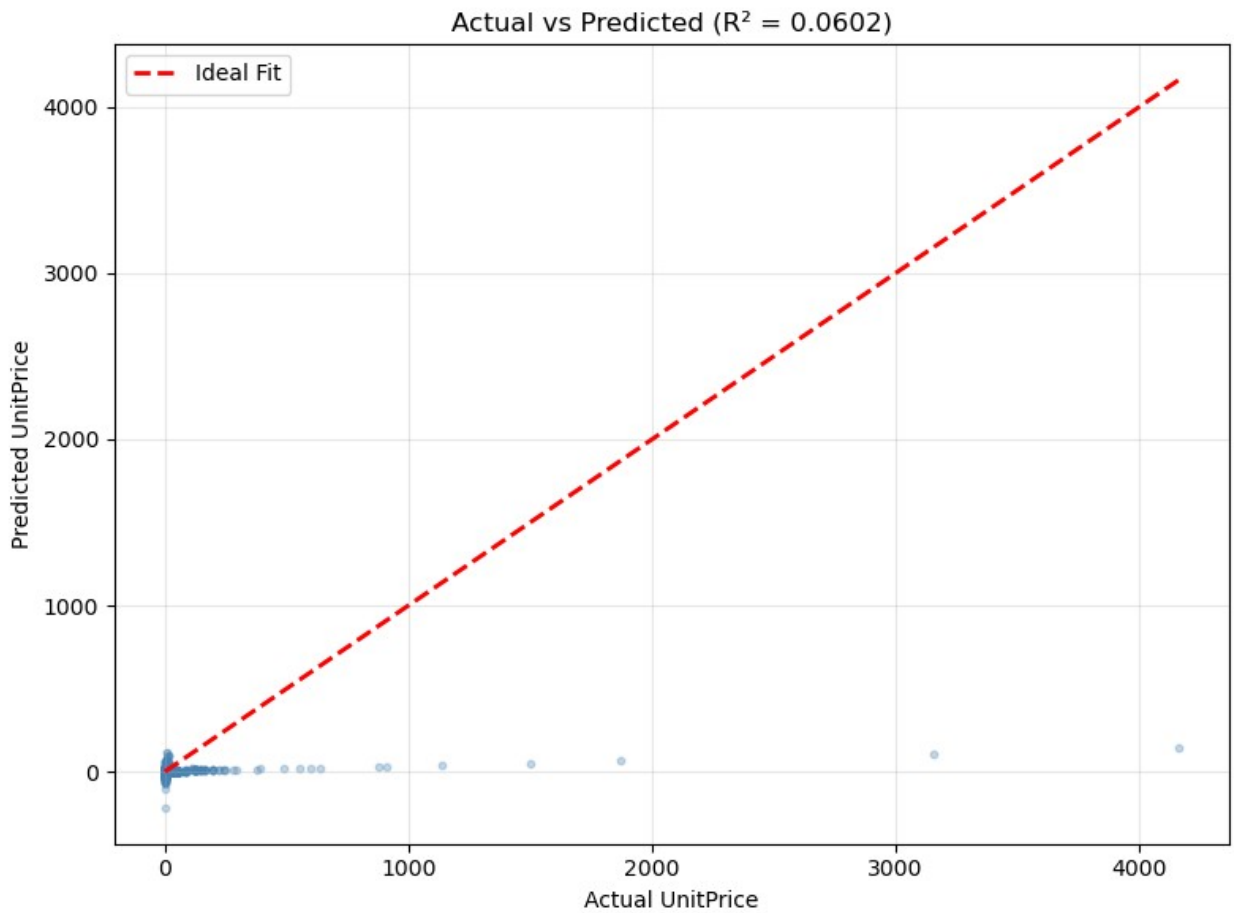
```
Training samples: 278518
Test samples    : 119366
```

```
# Compute All Regression Metrics

# 1. Pearson Correlation Coefficient (r)
r, p_value = pearsonr(y_test_r, y_pred_r)

# 2. Coefficient of Determination (R²)
r2 = r2_score(y_test_r, y_pred_r)

# 3. Mean Squared Error (MSE)
mse = mean_squared_error(y_test_r, y_pred_r)

# 4. Root Mean Squared Error (RMSE)
rmse = np.sqrt(mse)

# 5. Root Mean Squared Relative Error (RMSRE)
rmsre = np.sqrt(np.mean(((y_test_r.values - y_pred_r) /
y_test_r.values) ** 2))

# 6. Mean Absolute Error (MAE)
mae = mean_absolute_error(y_test_r, y_pred_r)
```

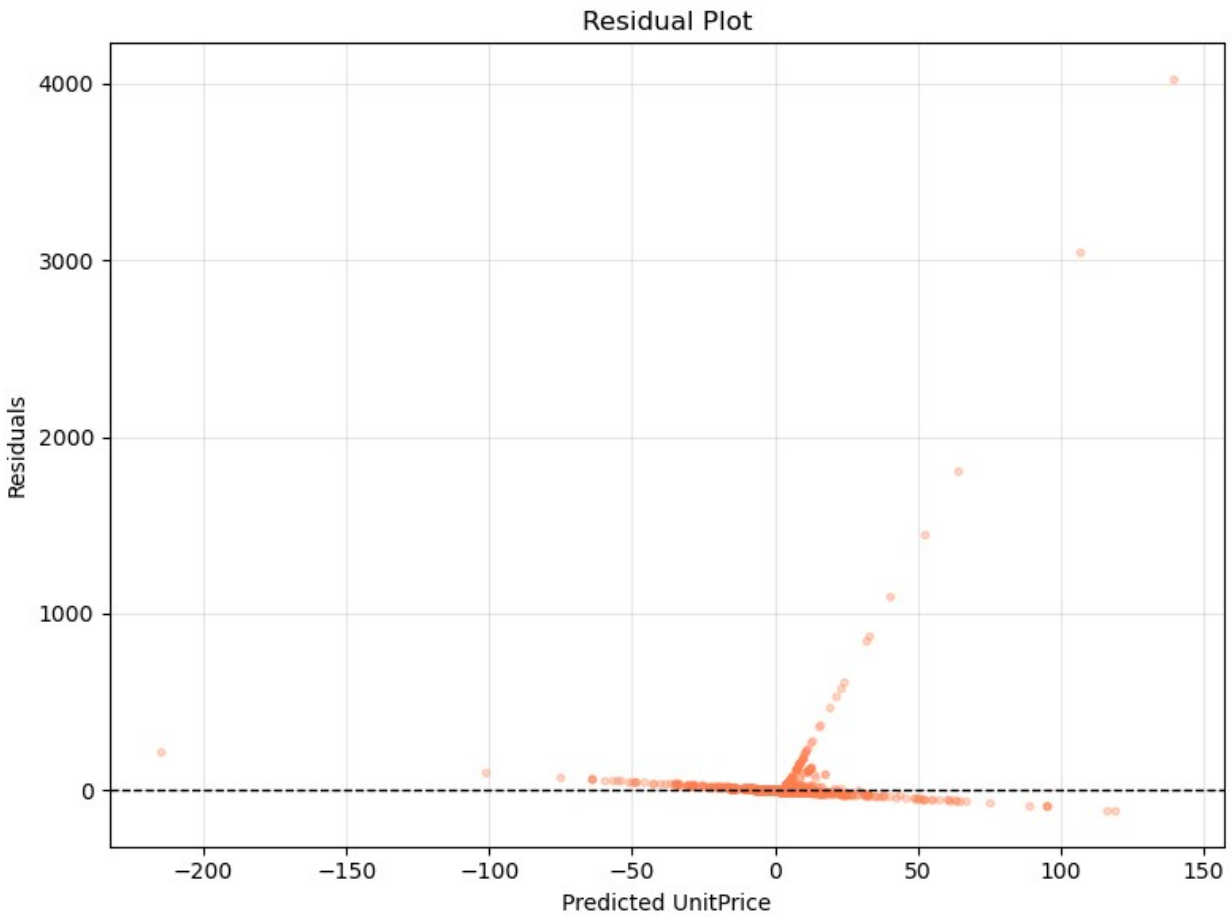
```python
# 7. Mean Absolute Percentage Error (MAPE)
mape = np.mean(np.abs((y_test_r.values - y_pred_r) / y_test_r.values))
* 100

print("=" * 50)
print("       REGRESSION EVALUATION METRICS")
print("=" * 50)
print(f"  Pearson Correlation (r) : {r:.4f}")
print(f"  R² Score                : {r2:.4f}")
print(f"  Mean Squared Error (MSE): {mse:.4f}")
print(f"  Root MSE (RMSE)         : {rmse:.4f}")
print(f"  Root MSR Error (RMSRE)  : {rmsre:.4f}")
print(f"  Mean Absolute Error(MAE): {mae:.4f}")
print(f"  MAPE                    : {mape:.2f}%")
print("=" * 50)
```

```
==================================================
       REGRESSION EVALUATION METRICS
==================================================
  Pearson Correlation (r) : 0.3255
  R² Score                : 0.0602
  Mean Squared Error (MSE): 314.5739
  Root MSE (RMSE)         : 17.7362
  Root MSR Error (RMSRE)  : 11.2167
  Mean Absolute Error(MAE): 2.1765
  MAPE                    : 159.02%
==================================================
```

```python
# Actual vs Predicted Plot
plt.figure(figsize=(8, 6))
plt.scatter(y_test_r, y_pred_r, alpha=0.3, s=10, color='steelblue')
plt.plot([y_test_r.min(), y_test_r.max()], [y_test_r.min(),
y_test_r.max()],
         'r--', lw=2, label='Ideal Fit')
plt.xlabel('Actual UnitPrice')
plt.ylabel('Predicted UnitPrice')
plt.title(f'Actual vs Predicted (R² = {r2:.4f})')
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```




```python
# Residual Plot
residuals = y_test_r.values - y_pred_r

plt.figure(figsize=(8, 6))
plt.scatter(y_pred_r, residuals, alpha=0.3, s=10, color='coral')
plt.axhline(y=0, color='black', linestyle='--', lw=1)
plt.xlabel('Predicted UnitPrice')
plt.ylabel('Residuals')
plt.title('Residual Plot')
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



Conclusion

In this experiment, we implemented and explored performance evaluation metrics for supervised learning models:

Classification (Logistic Regression):

- Computed Accuracy, Error Rate, Precision, Sensitivity, Specificity, F_1 Score, Geometric Mean, FPR, FNR, and ROC-AUC from the confusion matrix.
- Plotted the ROC curve to visualize classifier performance.

Regression (Linear Regression):

- Computed Pearson Correlation (r), R^2, MSE, RMSE, RMSRE, MAE, and MAPE.
- Visualized Actual vs Predicted values and Residuals.